KnowSelf

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Agentic knowledgeable self-awareness for planning systems

Source: arxiv.org/abs/2504.03553

Audience: ML researchers & graduate students

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Current Agent Training Uses 'Flood Irrigation'

Problem: indiscriminate knowledge injection during training/inference

Trajectory Imitation

- Learns from gold trajectories
- Limited adaptability
- No situation awareness

Trial-and-Error

- Reflects from failures
- High token/time cost
- Cannot know when to stop

Knowledge-Augmented

- Injects external knowledge
- Often applied everywhere
- Wasteful and sometimes harmful

Human decision-makers allocate effort selectively; current agents do not. KnowSelf introduces situational self-awareness to regulate cognition and knowledge use.

Three Situational Modes: Fast / Slow / Knowledgeable Thinking

Decision criterion: can the agent produce a correct next action immediately?

State + Task + Current Policy

Fast Thinking

Directly predicts correct action.
No reflection or retrieval needed.

Slow Thinking

Initial action uncertain/wrong.
Self-reflection revises decision.

Knowledgeable Thinking

Task requires external knowledge.
Trigger retrieval before acting.

Outputs are marked via special tokens to learn when to think, reflect, or seek knowledge.

KnowSelf Pipeline: Data Construction → Training → Self-Aware Inference

1) Data Construction

Self-explore ALFWorld trajectories, then heuristic situation judgment labels each step with mode tokens (<FAST>, <SLOW>, <KNOW>) based on action correctness and knowledge need.

2) Two-Stage Training

Stage A: supervised fine-tuning on token-annotated trajectories. Stage B: reasoning preference optimization (RPO) to reinforce better cognitive routing decisions.

3) Self-Aware Inference

At runtime, model emits mode token first. <SLOW> triggers reflection; <KNOW> triggers retrieval; <FAST> executes directly, reducing unnecessary external calls.

Framework reference: Figure 3 (data construction, two-stage training, self-aware inference).

Qualitative ALFWorld Cases: Reflection and Knowledge Triggering

Task A: Put a hot mug in cabinet

KnowSelf

KnowSelf route: <SLOW> reflection → heat mug → place in cabinet

Baseline

Baseline (OpenAI-O1): misses heating prerequisite, task fails.

Task B: Put saltshaker in drawer

KnowSelf

KnowSelf route: <KNOW> retrieval → locate valid container → place successfully

Baseline

Baseline (DeepSeek-R1): gets stuck in suboptimal action loop.

Figure 1 examples show KnowSelf's self-awareness resolves hidden preconditions and avoids dead-end action loops.

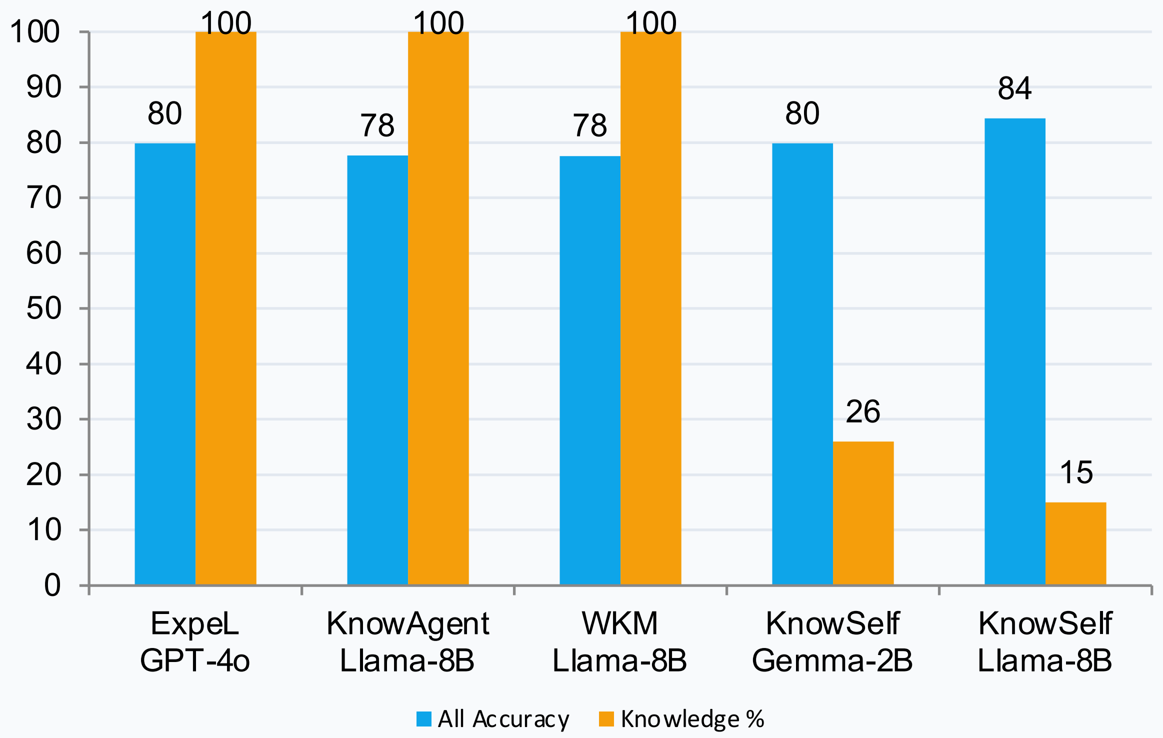
ALFWorld Results: KnowSelf Leads with Selective Knowledge Usage

All = average over six task types (Pick, Clean, Heat, Cool, Examine, Put).

Method	Backbone	Pick	Clean	Heat	Cool	Examine	Put	All	Know%
ReAct	GPT-4o	78.57	85.37	66.67	95.24	41.18	89.71	76.12	0%
Reflexion	GPT-4o	82.14	87.80	66.67	95.24	52.94	89.71	79.08	0%
ExpeL	GPT-4o	89.29	90.24	71.43	95.24	52.94	79.41	79.85	100%
ReAct	Llama-8B	83.93	85.37	57.14	90.48	47.06	79.41	73.90	0%
ETO	Llama-8B	89.29	90.24	71.43	95.24	52.94	79.41	79.85	0%
KnowAgent	Llama-8B	83.93	87.80	71.43	90.48	58.82	73.53	77.67	100%
WKM	Llama-8B	80.36	87.80	66.67	95.24	52.94	82.35	77.56	100%
KnowSelf	**Llama-8B**	**92.86**	**95.12**	**80.95**	**95.24**	**58.82**	**82.35**	**84.33**	**15%**
ReAct	Gemma-2B	83.93	90.24	57.14	95.24	47.06	73.53	74.52	0%
ETO	Gemma-2B	89.29	85.37	66.67	95.24	47.06	73.53	76.19	0%
KnowAgent	Gemma-2B	80.36	87.80	66.67	95.24	58.82	67.65	76.09	100%
WKM	Gemma-2B	76.79	85.37	73.81	95.24	58.82	67.65	76.28	100%
KnowSelf	**Gemma-2B**	**89.29**	**92.68**	**71.43**	**95.24**	**64.71**	**76.47**	**79.85**	**26%**

Best result: KnowSelf (Llama-8B) = 84.33 All at only 15% knowledge usage.

Efficiency Advantage: Lower Knowledge Use, Higher Accuracy



Key efficiency signals

- KnowSelf uses 15–26% knowledge
- Full-knowledge baselines use 100%
- KnowSelf-Llama-8B still has best All (84.33)
- Gemma-2B KnowSelf matches GPT-4o ExpeL (79.85)

Selective retrieval > blanket retrieval

Key Takeaways and Future Directions

Self-awareness is the missing control layer in LLM agent planning.

1. Indiscriminate knowledge injection is inefficient and can reduce planning performance.
2. KnowSelf learns situational routing with lightweight heuristic labels and special tokens.
3. Llama-8B KnowSelf outperforms GPT-4o ReAct (84.33 vs 76.12) and GPT-4o ExpeL (79.85).
4. Only 15–26% knowledge usage is required versus 100% for full-knowledge alternatives.
5. The paradigm scales: Gemma-2B KnowSelf reaches 79.85, matching GPT-4o ExpeL.

Future directions: transfer beyond ALFWorld, richer uncertainty calibration, and multi-agent coordination policies.